

Jose Bande Serrano

Full-stack Software Engineer

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Professional Summary

Full-stack Software Engineer with 12+ years of experience designing and delivering cloud-native, data-driven, and AI-powered applications across frontend, backend, data, and automation systems.

Selected Skills

Authentication, Azure, Bootstrap, CSS, Containerization, Database Design, Docker, Docker Compose, ESLint, Express, GitHub Actions, Go, GraphQL, HTML, Infrastructure as Code, JSON, JavaScript, Jest, Less, Next.js, Nuxt.js, PHP, Playwright, Prettier

Professional Experience

NBCUniversal - Full-Stack Engineer, Compliance Automation

2024-09 - present | New York, NY

Built and delivered a production compliance automation platform combining a high-performance Next.js SPA, backend workflows, ETLs, and reporting systems.

- Implemented a high-performance SPA in Next.js and TypeScript with TanStack-based data fetching and state management.
- Architected backend workflows, ETLs, and database design for reporting and control use cases.
- Delivered a production system used by thousands of NBCUniversal employees and reduced manual work by up to 80%.

Enerex - AI Development Consultant

2023-11 - present

Architected frontend and AI service layers for task-based automation systems and cross-product integrations.

- Architected and developed a React application with hybrid server/client rendering using TypeScript.
- Architected AI services as a multi-agent task-based automation system using Node.js, Azure Functions, Storage Queues, and SignalR.
- Improved throughput and developer experience by centralizing AI workflows and monitoring capabilities.

InteroCloud - Senior Full-Stack Developer

2022-11 - 2023-11

Contributed to a medical-practice CRM with Next.js SSR, Playwright automation, SEO improvements, and Python migration tooling.

- Contributed to a medical-practice CRM built with Next.js SSR and TypeScript.
- Implemented test automation with Playwright and added SEO-friendly pages.
- Introduced Google Tag Manager for reporting and event tracking.
- Built a document template generator that reduced support needs and saved internal development time.

Broadcasting Platforms - Senior Frontend Engineer & Team Lead

2019-11 - 2022-12

Led and built real-time streaming web applications for performers and end-users using modern frontend, backend jobs, and event-driven systems.

- Built real-time streaming web applications for performers and end-users using React with SSR on Next.js.
- Optimized site performance with asset compression, WebP conversion pipelines, and adaptive streaming support.
- Implemented Azure-based video processing functions and Python backend jobs.
- Architected chat, presence, notifications, and digital tipping flows using WebSockets and event-driven patterns.

First Due - Senior Software Engineer

2017-07 - 2019-11

Built large-scale web crawlers and ETL pipelines with Python and event-driven backend systems.

- Developed dozens of Python web crawlers and ETL pipelines to process large datasets into PostgreSQL.
- Implemented event-driven architecture with Django, Flask, RabbitMQ, and Scrapy for resilient extraction and scheduling.
- Deployed Elasticsearch for fast search across large datasets.

Liberty Power - .NET/SQL Consultant

2016-05 - 2017-06

Modernized legacy .NET applications and improved database quality and performance with SQL work.

- Migrated legacy VB/ASP.NET applications to ASP.NET with C#, Razor, and Entity Framework.
- Authored SQL stored procedures and data-fix scripts to improve data quality and performance.

DATYS - Computer Science Specialist

2009-09 - 2016-04

Worked on FPGA research, automation tooling, and distributed systems for fraud detection and identification platforms.

- Conducted research and built FPGA architectures for high-throughput network intrusion detection and published five peer-reviewed papers.
- Developed Python tooling to auto-generate FPGA designs and speed development cycles.
- Contributed to a nation-wide AFIS system and a telephone fraud detection system using distributed and parallel computing.

Projects

AI Assisted Invoice Processor

An application to help finance teams collect invoices, extract structured data, review exceptions, and move approved records into downstream reporting without juggling disconnected inboxes, sheets, and shared folders.

- Built for finance workflow automation around invoice collection, extraction, review, and downstream reporting.
- Public project URL: <https://invo.gnosiscraft.com>